Example 2.13 Find the length of the rope by which a cow must be tethered in order that it may be able to graze an area of 9856 *sq.m* (use $\pi = \frac{22}{7}$)

Solution

Given that, the area of the circle $= 9856$ *sq.m*

$$\pi r^2 = 9856$$

$$\frac{22}{7} \times r^2 = 9856$$

$$r^2 = 9856 \times \frac{7}{22}$$

$$r^2 = 448 \times 7 = 3136$$

$$r^2 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 7 \times 7$$

$$r^2 = 8 \times 8 \times 7 \times 7 = 8^2 \times 7^2 = (8 \times 7)^2$$

$$r = 8 \times 7 = 56 \ m$$

2	3136
2	1568
2	784
2	392
2	196
2	98
7	49
	7

Therefore the length of the rope should be 56 *m*.

Example 2.14 A garden is made up of a rectangular portion and two semicircular regions on either sides. If the length and width of the rectangular portion are 16 *m* and 8 *m* respectively, calculate $(\pi = 3.14)$



Fig. 2.21

(i) the perimeter of the garden (ii) the total area of the garden

Solution

(i) The perimeter of the rectangular garden

Perimeter include two lengths each of 16 *m* and two semi circular arcs of diameter 8 *m*.

$$\text{Circumference of the semicircle} = \frac{\pi d}{2} \ units$$

$$= \frac{\pi \times 8}{2} = 4\pi$$

$$= 4 \times 3.14$$

$$= 12.56 \ m$$

Therefore, circumference of two semicircles $= 2 \times 12.56$

$$= 25.12 \ m$$

Total perimeter = length + length + circumference of two semicircles

$$= 16 + 16 + 25.12$$

$$= 32 + 25.12$$

$$= 57.12 \ m$$

(ii) The total area of the garden

Total area of the garden = Area of rectangle + Area of 2 semicircles

= Area of rectangle + Area of the circle

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